CL 299 presentation 2

LNP Manufacturing and Process Simulation

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Quenching

Quenching is the immediate dilution of the freshly formed LNP suspension with an excess aqueous buffer.

- First , LNPs are formed.
- Now, quench buffer is added which results in sharp drop of ethanol concentration which reduces lipid solubility and mobility.
- Finally, particle growth ceases, preventing further fusion or aggregation, thereby locking in the final particle size and PDI.
- The quenched stream exits the system, and the LNPs are collected in a sterile container or bag.

Down-Stream Process

Downstream processing includes all steps carried out after product formation to purify, stabilize, ensure safety, and make it suitable for storage.

- After LNP formation, the suspension still contains residual ethanol, acidic buffer, free lipids, unencapsulated mRNA, incorrect concentration, and lacks sterility. Thus, while the LNP structure is correct, the surrounding environment and purity are unsuitable for use.

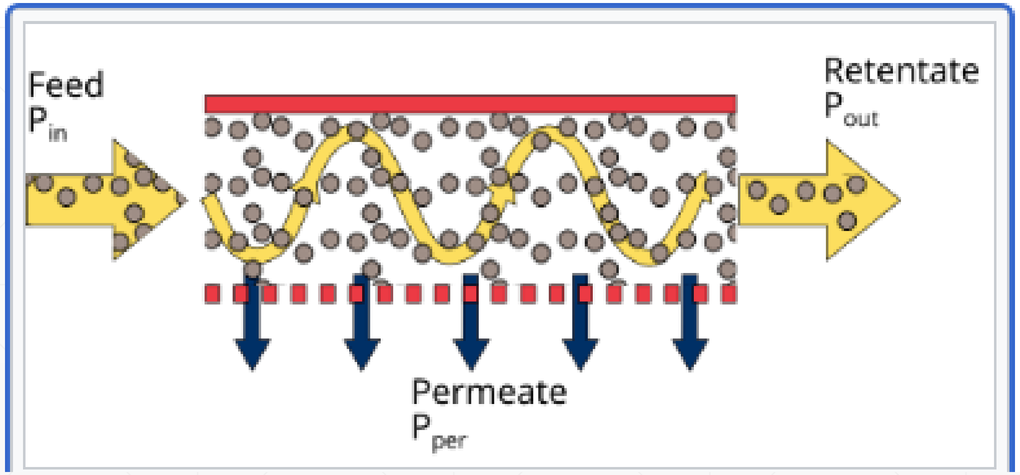
- Downstream processing ensures removal of ethanol and impurities, addition of buffer, and dosage adjustment, and a stable, sterile product for storage.

- To Achieve this we use Tangential Flow Filtration(TFF), Sterile Filtration ,Fill–Finish.

Tangential Flow Filtration (TFF)

TFF simultaneously removes ethanol, enables buffer exchange (pH adjustment), and controls product concentration (Core purification & conditioning step).

- LNP suspension flows **parallel** to a membrane.
- Small molecules (ethanol, salts) pass through MWCO memberance in which driving force is TMP.
- LNPs are retained and recirculated.
- Fresh buffer is continuously added (diafiltration).



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Membrane MWCO

MWCO stands for Molecular Weight Cut-Off. It is the approximate molecular weight of the smallest molecule that the membrane will retain.

- MWCO is not an exact pore diameter. It is an operational definition, based on: Shape, Flexibility, Hydration of molecules.

- Ethanol, free lipid, M-RNA, salts, Buffer-molecule have approx. (MW :- /1KDa) which can easily pass through MWCO (100–250 KDa).

LNP = MW > 500KDa

- Molecules smaller than MWCO → pass through the membrane

- Molecules larger than MWCO → are retained

Transmembrane Pressure (TMP)

- TMP is the pressure difference across the membrane. It is the driving force pushing small molecule to pass through the membrane.
- It Determines permeate flow rate, Controls how fast ethanol and buffer are removed.
- If TMP is too low then , Slow filtration and Long processing time.
- If TMP is too high then, Membrane fouling and LNP deformation or rupture.
- ***Cross-flow Rate(tangential velocity)*** *is speed at which the LNP suspension flows parallel to the membrane.*
- ***Diafiltration volume*** *defines how many times the LNP solution is washed with fresh buffer during TFF, determining how completely small impurities are removed.*

Alternatives to TFF

Alternative	When Used	Limitations
Dialysis	Lab Scale only	Very Slow, Not scalable
Centrifugal purifications	Small Batches	Shear, Low scalablity
SEC(Size Exlusions)	Analytical Purifications	Not Industrial
Precipitations	Rare	High LNP loss

Sterile Filtration

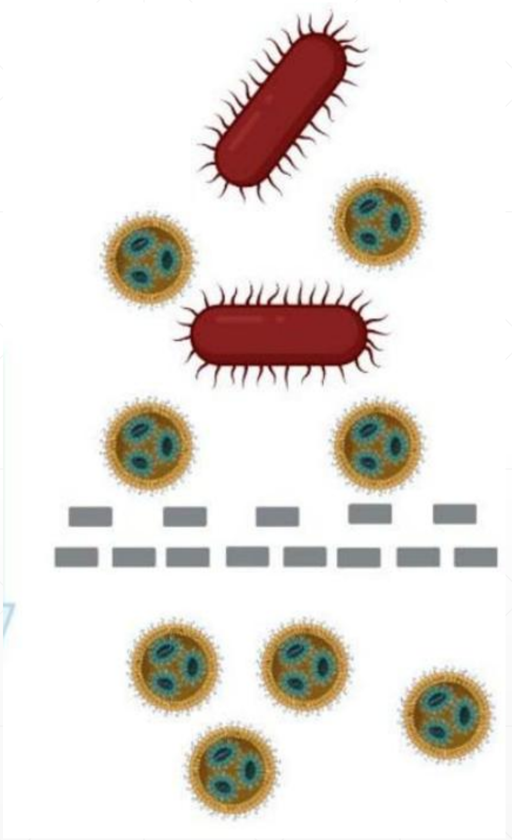
- Sterile filtration is a physical separation process used to remove microorganisms from the LNP formulation without using heat or chemicals.

- LNPs are injectable products which must be free from bacteria.It removes bacteria using a 0.22 µm filter.

It is done by :-

- Low-pressure filtration

- Lipid-compatible membranes



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Alternative Method	Why not used in LNP Formation
Heat Sterilization	Destroy M-RNA and melt lipid structure
Gamma irradiation	Cause RNA strands to break
Chemical sterilants	Leaves toxic residues
0.45 µm filtration	Not sufficient to ensure sterility

- Sterile filtration is the only practical sterilization method for mRNA-LNPs.

Fill–Finish

- It is needed to Accurate dosing and Protection from contamination.
- First we do Aseptic method to fill in Vials and Syringes.
- Then, Stoppering and sealing and at last Labeling and batch tracking.

Storage

- It is stored between –80°C to –20°C.
- Must be Stored in RNase Free Environment.

Source:https://www.ubrigene.com/wp-content/uploads/2024/10/Need-Aseptic-Fill-Finish-Support.jpg

mRNA Feed (Aqueous Phase)

- Single standed messenger RNA is dissolved in aqueous buffer.
- Buffer Concentration ≤ 25 mM.
- Typically 0.05 – 1 mg/mL
- Chosen based on desired final dose & encapsulation efficiency
- pH = 4.0 during mixing

Buffer Systems and Lipid-RNA Relationships

- **Allowed buffer bases:**
- Tris, Bis-Tris, Triethanolamine (TEA)
- **Allowed counter-anions:**
- Chloride, Acetate, Lactate, Glycolate, etc.
- Buffer Concentration $\leq$ 25 mM
- Lipid : RNA (mass ratio):
  - ~9 : 1 to ~20 : 1

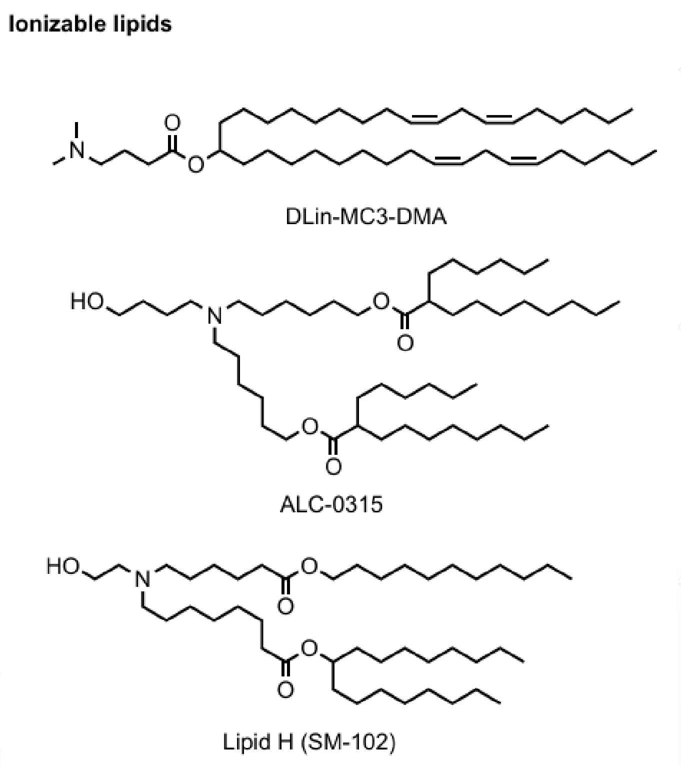
Lipid Feed (Organic Phase)

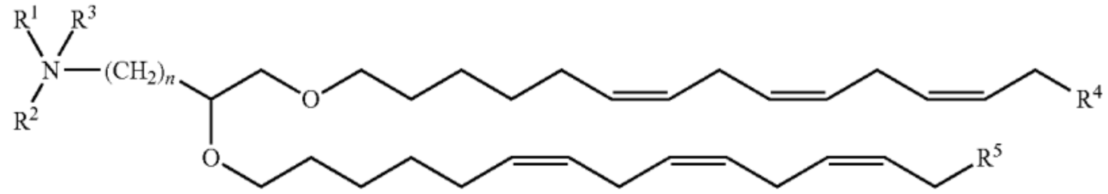
Component	Typical Role
Ionizable lipid	Electrostatic binding to mRNA
Phospholipid	Structural scaffold
Cholesterol	Membrane rigidity & packing
PEG-lipid	Size control & colloidal stability

Total lipid concentration is 10 mM

Ionizable Lipids

- There are large libraries of ionizable/cationic lipids, including:
- Tertiary amine–based ionizable lipids
- Biodegradable ionizable lipids
- Ester-, amide-, and carbamate-linked lipids
- Examples: DLin-MC3-DMA, SM-102, ALC-0315, C12-200
- Key property: pKa ≈ 6.2 – 6.8
- pH 4 ensures full protonation of ionizable lipid

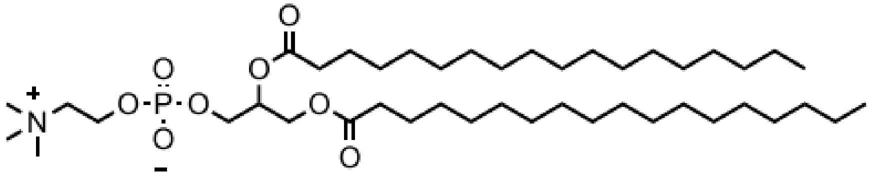




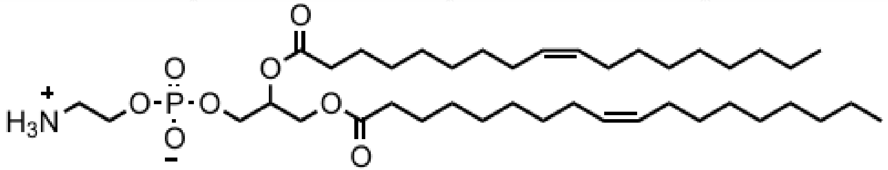
General structure of cationic lipids

Helper Lipids Examples

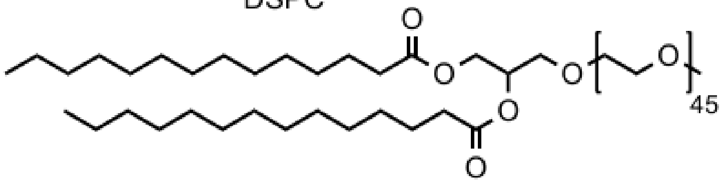
- DSPC, DPPC, DMPC, ALC-0159, DOPC, DOPE, DOPG, DOPS
- Sphingomyelins, Glycolipids
- **Role:**
- Structural stability and membrane organization



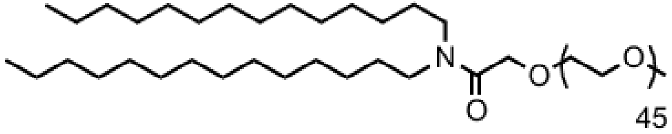
DSPC



DOPE



PEG_{2000}-DMG



ALC-0159

Sterols & Lipopolymers

- **Sterols:**
- Cholesterol
- Derivatives
- Phytosterols
- Increase membrane rigidity
- **Lipopolymers:**
- PEG-lipids
- PEG-phospholipids
- Polysarcosine-lipids
- Prevent aggregation

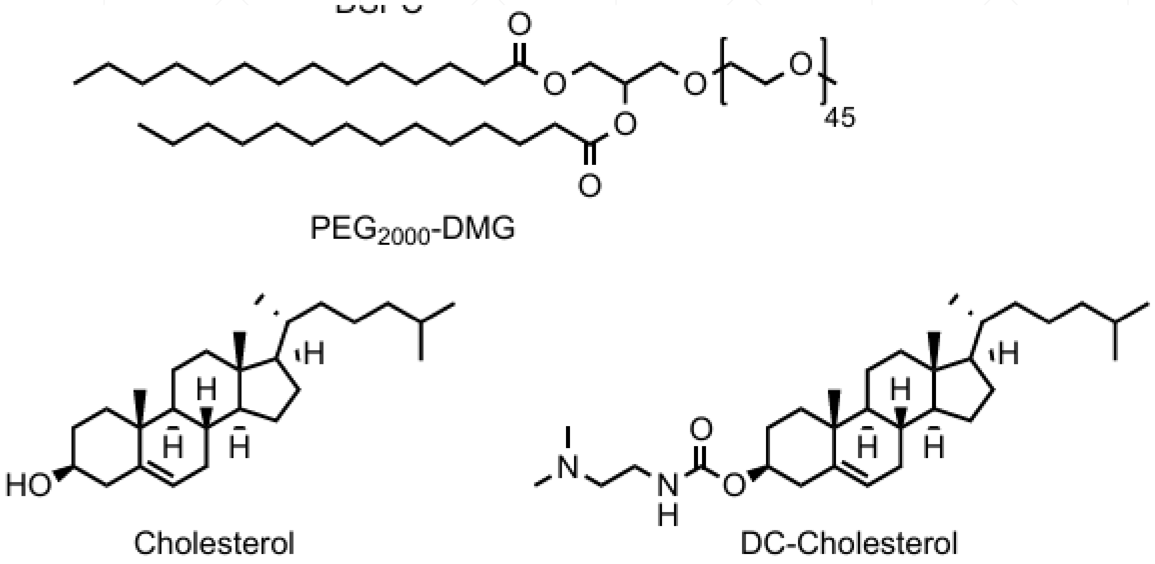


Table 1: *Theoretical molar composition of C12-200/SM-102- and MMG-1-modified C12-200/SM-102 lipid nanoparticles (LNPs)*

Formulations	Composition (mol %)				
	C12-200	DOPE	Cholester ol	MMG-1	DMPE-PEG_{2000}
C12-200	35	16	47.5	0	1.5
C12-200 MMG 11	35	16	35.63	11.88	1.5
C12-200 MMG 23	35	16	23.75	23.75	1.5
C12-200 MMG 35	35	16	11.88	35.63	1.5
C12-200 MMG 47	35	16	0	47.5	1.5
C12-200 DOPE DMPE-PEG2000	66.65	30.48	0	0	2.87
	SM-102	DSPC	Cholester ol	MMG-1	DMG-PEG_{2000}
SM-102	50	10	38.5	0	1.5
SM-102 DSPC MMG 37	50	10	0	37.5	2.5
	SM-102	DOPE	Cholester ol	MMG-1	DMG-PEG_{2000}
SM-102 DOPE MMG 38	50	10	0	38.5	1.5

An example of composition from EPO patent. C12-200 LNPs were prepared by microfluidic mixing using the NanoAssemblr® Ignite™ microfluidic mixer system (Precision Nanosystems Inc., Vancouver, Canada), which uses a toroidal micromixer chip.

MMG: monomycoloyl glycerol

Design Space

Component	Allowed Range (mol%)	Preferred Range
Ionizable lipid	10 – 60	30 – 40
Helper lipid	5 – 30	10 – 20
Sterol	5 – 70	10 – 50
Lipopolymer	0.5 – 50	1 – 5

Aqueous : Organic = 1:1 to 3:1, with 3:1 most commonly reported.

Obtaining structurally distinct LNPs often requires drastically changing the chemical composition.

N/P ratio (critical feed parameter)

- **N/P ratio** =
  (Number of protonatable nitrogens in ionizable lipid) / (Number of phosphate groups in RNA)
- Values used: mRNA → N/P = 6

  siRNA → N/P = 3
- Controls how much lipid is required per RNA
- Directly affects:
  - particle formation
  - charge balance
  - encapsulation

Feed Variables Classified (Engineering View)

- **Fixed / chosen upfront:**
- Lipid identity
- Lipid molar ratio
- RNA concentration
- Buffer pH

- **Tunable during operation:**
- Flow rate
- Flow ratio
- Mixing intensity

- This classification is excellent for simulation justification.

Process Analytical Technology(PAT)

- FDA defines PAT as a system to analyse, monitor, and control pharmaceutical manufacturing processes in real time.
- It is used for taking measurements directly during the process.
- PAT data may measure chemical or physical aspects.

Table 1. Basic Contrast between PAT in Development and Manufacturing

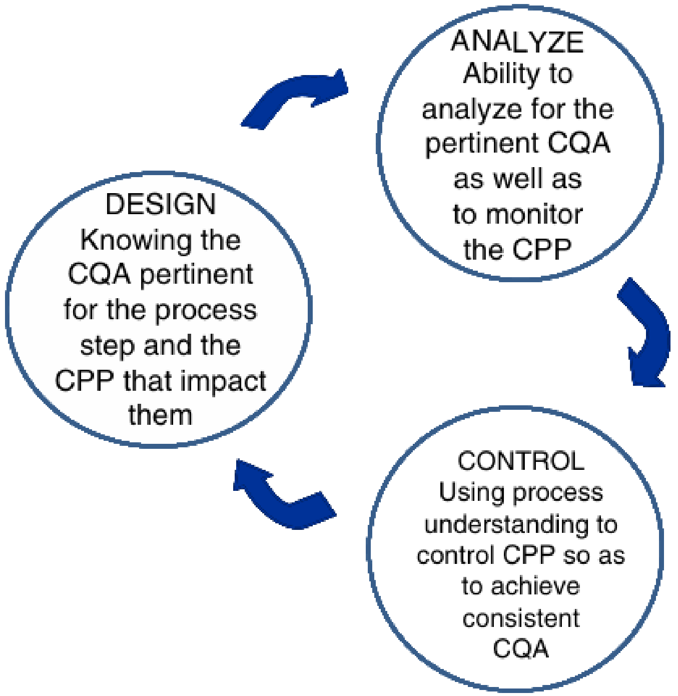
	Development	Manufacturing
Overall Purpose	Understand	Control, trend analysis
Desired Technology	Multicomponent analyzers	Targeted analyzers
Data Complexity	Multivariate	Univariate or multivariate[a]
Support Requirements	High level expertise—continuous support	Robust and automated—minimal support
Quality System	Development mode	GMP
PAT Expertise	Method design and development	Operation and maintenance

Why Use PAT?

- Real-time and frequent measurements.
- Improved process understanding.
- Better process control and consistency.
- Enhanced safety.
- Ability to monitor difficult systems.
- Improved scale-up and technology transfer.

PAT Implementation

- PAT integrates process understanding, real-time analysis, and control to ensure product quality.



The three steps that must be taken for PAT implementation,

Role of PAT in LNP Manufacturing

- Lipid nanoparticles are highly sensitive to process conditions
- Small process variations can significantly affect:
  - Particle size
  - Stability
  - Drug delivery performance
- PAT enables real-time monitoring and control during LNP production
- It focuses on identifying and controlling:
  Critical Quality Attributes (CQAs)

  Critical Process Parameters (CPPs)

Some Analytical techniques used in PAT

- **Dynamic Light Scattering (DLS)**
  This technique is used to measure particle size and size distribution in real-time.
- **Zeta Potential Analysis**
  This technique determines the surface charge of LNPs.
- **X-ray Diffraction (XRD):**
  Analyzes the crystalline or amorphous nature of lipid components within LNPs.
- **Mass Spectrometry (MS):**
  Provides information on molecular weight and composition of lipids, especially ionizable lipids.
- **Nuclear Magnetic Resonance (NMR):**
  Provides detailed insights into lipid structure, composition, and molecular interactions in LNP formulations.

- **High-Performance Liquid Chromatography (HPLC):**
  Quantifies lipid components and encapsulated payload (e.g., mRNA, siRNA).

- **Differential Scanning Calorimetry (DSC):**
  Used to study the thermal behavior and phase transitions of lipids in LNP formulations.

PAT Tools

- Multivariate tools for design, data acquisition, and analysis
- Process analyzers
- Process control tools
- Continuous improvement and knowledge management tools

An appropriate combination of some or all of these tools may be applicable to a single unit operation or to an entire manufacturing process and its quality assurance.

PAT Measurement Level

- describe how closely an analytical method is connected to the manufacturing process and how useful it is for real-time control.
- Four Analytical levels:

  Offline: analysis after sample removal

  At-line: analysis near the manufacturing line

  In-line: sensor placed directly in the process stream

  Online: automated sampling and analysis loop
- Process relevance and control capability increase from offline to online.

Conclusion

- Process Analytical Technology (PAT) helps monitor and control pharmaceutical processes while manufacturing is happening.
- Instead of checking quality only at the end, PAT checks quality during the process, which improves reliability.
- Different measurement approaches (offline, at-line, inline, and online) allow PAT to be used flexibly in many processes.
- In lipid nanoparticle (LNP) manufacturing, PAT is very important because it helps control particle size, which affects product performance.
- Overall, PAT supports better process design, consistent quality, and modern manufacturing methods such as continuous processing.

Resources:

- https://www.sciencedirect.com/science/article/abs/pii/S0378517325003576
- https://pubmed.ncbi.nlm.nih.gov/40139451/
- https://www.cytivalifesciences.com/en/us/insights/filtration-of-nucleic-acids-and-lnps-in-mrna-manufacture
- https://www.sharpservices.com/expert-content/beyond-lipids-the-science-of-lnp-manufacturing-and-fill-finish/
- https://advanced.onlinelibrary.wiley.com/doi/10.1002/adma.202419538
- https://www.researchgate.net/publication/361170223_Lipid_nanoparticle_formulations_for_optimal_RNA-based_topical_delivery_to_murine_airways
- https://data.epo.org/publication-server/rest/v1.2/patents/EP4442256NWA1/document.html
- https://patentimages.storage.googleapis.com/4b/d7/bf/442f9a59aba04c/US11191849.pdf
- https://patentimages.storage.googleapis.com/b9/d6/94/1da620d0bbb5a7/US20230414747A1.pdf

- https://pubs.acs.org/doi/full/10.1021/acs.accounts.1c00544
- https://www.tandfonline.com/doi/full/10.1517/17425247.2012.673278#d1e579
- https://www.nature.com/articles/s41578-021-00358-0
- https://www.tandfonline.com/doi/full/10.2147/IJN.S123062#d1e182
- https://link.springer.com/article/10.1007/s00216-010-3781-x#Sec8
- https://pubs.rsc.org/en/content/articlehtml/2010/ay/c0ay00257g
- https://pubs.acs.org/doi/full/10.1021/op400358b

THANK YOU!!!